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Proposed Project Title 1: Capstone Coffee Co. Database Management System	

1. Area of Investigation:

The study explores the design and implementation of a Database Management System for a Coffee Shop. The aim of the system is to arrange important business data, including suppliers' information, sales transactions, and product inventories. The system will act as a repository, ensuring that all business activity is effectively documented, retrieved, and preserved.

1.1 Developing a relational database structure that supports efficient data normalization, indexing, and retrieval.

2. Background and Rationale of the Project/System

A lot of coffee shop owners, especially the small coffee shops, are relying on manual record keeping to track their daily sales, inventories, and other information. These methods often lead to human error, such as inaccurate inventory counts and data loss. This Coffee Shop Database Management System aims to address the challenges by providing a well-structured relational database. This system will improve data accuracy.

Coffee Shop Database offers:

- 2.1 Automate sales and inventory tracking to minimize human error.
- 2.2 Quickly generate reports on best-selling products and customers' orders.
- 2.3 Securely store and retrieve employee and suppliers' information.
- 2.4 Maintain consistent and accessible data for management use.

3. Statement of Objective

General Objective

- To develop a Database Management System for Capstone Coffee Co. that efficiently stores, organizes, and retrieves data related to sales, inventory, suppliers, and employees to improve business operations and accuracy.

Specific Objective

- 3.1 Design a relational database that systematically organizes specific and precise records of sales, inventories, and suppliers.
- 3.2 Implement data automation features to simplify record management and minimize manual intervention.
- 3.3 Establish secure access controls to safeguard confidential business and employees' information.
- 3.4 Generate detailed analytical reports to support data-driven management decisions.
- 3.5 Ensure system scalability and flexibility to adapt to future business expansion or operational needs.

4. Scope and Limitations of the Study:

Scope:

The main goal of the system is to improve data accuracy and consistency by moving the company from manual record-keeping to a well-structured relational database solution. In terms of functionality, the study discusses the architecture required to effectively store, arrange, and retrieve essential business data, such as all sales transactions, product inventories, and supplier records. The final system will be designed with a relational structure that supports effective data normalization, indexing, and quick retrieval for management use. It will also automate data recording and report generation to make monitoring easier.

Limitations:

The Database does not handle online ordering, mobile app integration, or automated payment processing. Its functionality relies on manual input of daily transactions, and it is limited to the internal operations of Capstone Coffee Co. External data sources or third-party integrations are not included in the current design.

5. Target Beneficiaries:

5.1 Coffee Shop Owners

- Reduce the time spent on making manual record tracking.

5.2 Employees

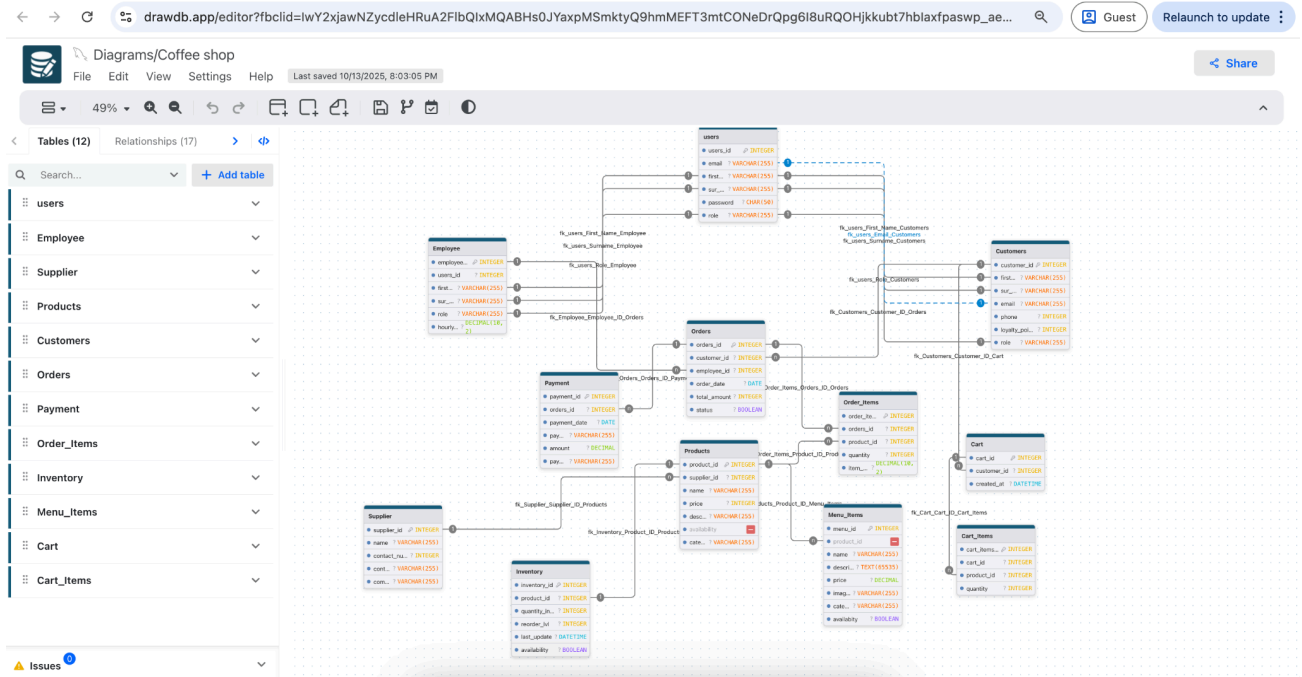
- Experience a more structured workflow through user-friendly data forms.

5.3 Programmers

- They will gain practical experience through their hands-on management of the database system.

5.4 Other Business Companies

- Not only coffee shop owners, but also other product-based businesses can benefit from it.



6. References:

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